

Ketogenic Diets and Depression and Anxiety

A Systematic Review and Meta-Analysis

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[+ Supplemental content](#)

IMPORTANCE Ketogenic diets (KDs) have been hypothesized to influence mental health through pathways involving mitochondrial function, inflammation, and neurotransmitters, but their therapeutic value in psychiatric populations remains uncertain.

OBJECTIVE To assess the associations between KDs and mental health outcomes in adults, with a focus on depressive and anxiety symptoms.

DATA SOURCES MEDLINE, Embase, and APA PsycINFO were searched on April 18, 2025. Additional studies were identified through manual searches and clinical trial registries.

STUDY SELECTION Studies involving adults aged 18 years or older receiving a KD (<26% energy from carbohydrates or <50 g/day) and assessed with validated psychiatric scales were eligible. Designs included randomized clinical trials (RCTs), quasi-experimental (QSE) studies, cross-sectional studies, case series, and case reports.

DATA EXTRACTION AND SYNTHESIS Data were extracted by 1 reviewer and verified by 2 others. Risk of bias was assessed using critical appraisal tools from the Joanna Briggs Institute. Random-effects meta-analyses were run separately for RCTs and QSEs.

MAIN OUTCOMES AND MEASURES The primary outcome was changes in psychiatric symptom severity measured by standardized scales, reported as standardized mean differences (SMDs) or standardized mean change using change scores (SMCCs).

RESULTS A total of 50 studies (41 718 participants) were included. Ten RCTs on KD for depressive symptoms vs control diets showed a significant association (SMD, -0.48 ; 95% CI, -0.87 to -0.10 ; $I^2 = 67.2\%$), with stronger associations in studies using ketone monitoring, nonobese participants, very low-carbohydrate interventions, and non-high-carbohydrate comparators. Nine RCTs on anxiety showed no significant association (SMD, -0.03 ; 95% CI, -0.18 to 0.12 ; $I^2 = 41\%$). In QSEs, 9 on depressive symptoms showed a consistent association (SMCC, -0.66 ; 95% CI, -0.83 to -0.50 ; $I^2 = 0\%$), and 6 on anxiety showed similar results (SMCC, -0.58 ; 95% CI, -0.81 to -0.36 ; $I^2 = 0\%$).

CONCLUSIONS AND RELEVANCE In this systematic review and meta-analysis, KDs were associated with modest improvements in depressive symptoms, particularly with biochemical ketosis verification, while anxiety evidence was inconclusive. Given heterogeneity, comparators, and short follow-up, well-powered trials with standardized, verified protocols, structured support, and prespecified outcomes are needed to confirm efficacy and durability.

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The ketogenic diet (KD), which is high in fat, low in carbohydrates, and moderate in protein, was introduced in the 1920s as a nonpharmacological therapy for refractory epilepsy.^{1,2} Sustained carbohydrate restriction induces nutritional ketosis, in which ketone bodies—primarily β -hydroxybutyrate (BHB), acetoacetate, and acetone—replace glucose as the brain's primary energy source.³ This metabolic shift affects mitochondrial function, oxidative stress, and inflammatory signaling.⁴⁻⁶

Interest in the psychiatric applications of KD is growing, particularly for disorders linked to metabolic and neurobiological dysfunction.^{7,8} Major depressive disorder (MDD), bipolar disorder (BD), and schizophrenia (SZ) are associated with mitochondrial impairment, insulin resistance, cerebral glucose hypometabolism, and systemic inflammation.⁷⁻⁹ Ketone bodies may modulate these processes by enhancing mitochondrial bioenergetics, reducing reactive oxygen species, regulating neurotransmission, and altering gut microbiota.^{4,8,10,11}

Mechanistic parallels with epilepsy further support KD in psychiatry.^{9,12} Both conditions involve excitatory-inhibitory imbalance, synaptic instability, and altered metabolic signaling.⁹ KDs may stabilize neural networks, modulate γ -aminobutyric acid (GABA) or glutamate activity, and mimic the pharmacodynamics of mood stabilizers.⁷ Their glucose-sparing effect allows glucose to be redirected to neurotransmitter synthesis and antioxidant defense.^{3,13}

Preliminary studies and case reports suggest KD may reduce psychiatric symptoms and improve functioning in serious mental illness.^{7,8} Reported benefits include improvements in mood, anxiety, cognition, weight, and quality of life, with possible reductions in psychotropic use. Findings are limited by heterogeneity in design, population, protocol, and outcomes, and some data come from studies in nonpsychiatric populations where mental health outcomes were secondary.¹⁴⁻¹⁶ Safety concerns remain for certain groups, including those with mitochondrial DNA deletions or those receiving polypharmacy.¹⁷

Given growing interest in nutritional psychiatry and the need for interventions addressing psychiatric and metabolic dysfunction, this systematic review and meta-analysis evaluates associations of KD with mental health outcomes in psychiatric and other medical populations. By summarizing study characteristics and pooled associations, this work aims to inform future research and clinical practice.

Methods

This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) reporting guidelines¹⁸ and was registered with PROSPERO (CRD42024577648).

Search Strategy

On April 18, 2025, a search was conducted in MEDLINE (Ovid interface, 1946-2025), APA PsycINFO (Ovid interface, 1806-2025), and Embase (Ovid interface, 1974-2025) using key-

Key Points

Question What are the associations between ketogenic diets (KDs) and mental health outcomes in adults, particularly depressive and anxiety symptoms?

Findings In this systematic review and meta-analysis of 50 studies, KDs were associated with small to moderate improvements in depressive symptoms in randomized clinical trials and with larger improvements in quasi-experimental studies. No significant associations were found for anxiety in randomized clinical trials.

Meaning KDs show potential associations with reduced depressive symptoms, but evidence for anxiety is inconclusive; larger, high-quality trials are needed to clarify effectiveness and generalizability.

words and subject headings on 2 core concepts: ketogenic or low-carbohydrate diets and mental health disorders. The full search strategy is available in the eMethods in Supplement 1. To identify ongoing clinical trials that have not yet been published, registered clinical trial data were obtained from ClinicalTrials.gov.

Eligibility Criteria

Articles were eligible if they included human participants aged 18 years or older from psychiatric and other medical populations who received a KD. Interventions included high-fat, low-carbohydrate, moderate-protein diets (eg, standard KD, Modified Atkins Diet) or other structured regimens with carbohydrate intake less than 50 g/day or less than 26% of energy. Studies had to assess mental health outcomes using at least 1 validated psychiatric scale (eg, Patient Health Questionnaire-9 [PHQ-9], Generalized Anxiety Disorder-7 [GAD-7]). Eligible designs were original, peer-reviewed quantitative studies—randomized clinical trials (RCTs), quasi-experimental (QSE) studies, cohort studies, cross-sectional (CS) studies, or case reports—published in English. Reviews, protocols without results, commentaries, editorials, and animal studies were excluded. During full-text screening, low-carbohydrate diets were assessed to confirm macronutrient composition consistent with KD.

Articles were eligible for meta-analysis if they reported sufficient data to calculate standardized mean differences (SMDs). For RCTs, this required means and SDs at the primary end point for both intervention and control groups using validated psychiatric scales. For QSE studies, means and SDs at baseline and posttreatment within the same group were required.

Data Screening and Extraction

Records were imported into Covidence, and duplicates were removed automatically. Two reviewers screened titles or abstracts and full texts, with discrepancies resolved by a third reviewer or by consensus. Data extraction was performed by 1 reviewer and verified by 2 others. Extracted variables included study details (author, year, country, design, target disorder, primary mental health domain), participant characteristics (sample size, mean age, sex), psychiatric outcome

measures, intervention and control details, adjuvant treatments, primary and secondary outcomes, adverse events, main results, and conclusions of the authors.

For registered clinical trials, the following data were extracted: ClinicalTrials.gov identifier, target disorder, study status (eg, not yet recruiting), intervention type, primary and secondary outcome measures, participant sex, and study country.

For studies eligible for meta-analysis, additional variables were extracted, including study identifier, sample type (clinical or nonclinical), psychiatric scale used (eg, GAD-7, PHQ-9), direction of improvement, and mean and SD values, at the primary end point for RCTs and at baseline and posttreatment for other study designs.

Quality Assessment

Risk of bias was assessed using the appropriate Joanna Briggs Institute (JBI) critical appraisal tools for each study type.¹⁹ Each criterion was rated as “yes,” “no,” “unclear,” or “not applicable.” As the JBI tools do not produce a composite score, inclusion was based on reviewer judgment of each study’s relevance to understanding KDs in mental health. Two independent reviewers conducted the risk of bias assessments, which were reviewed by a third reviewer.

Statistical Analysis

Result Synthesis (Qualitative)

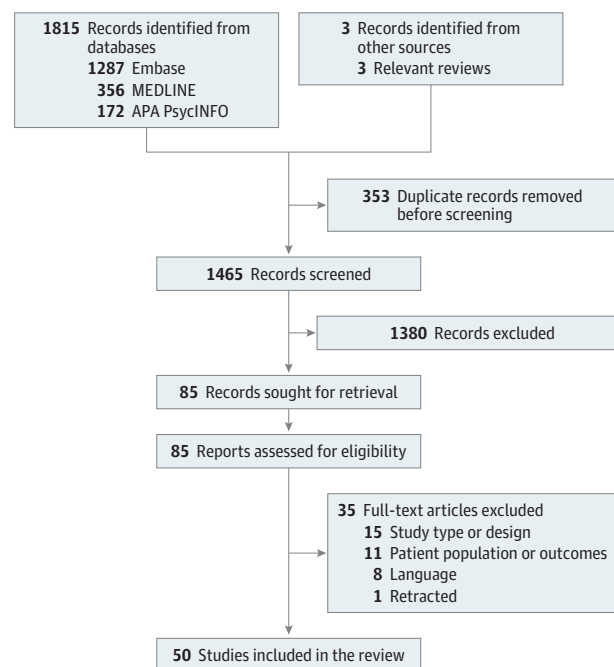
A narrative synthesis was conducted according to synthesis without meta-analysis (SWiM) guidelines,²⁰ focusing on mental health outcomes in psychiatric and other health-related populations. Studies were grouped by psychiatric domain, and standardized outcome measures summarized intervention associations. Symptom changes, intervention features, safety data, and between-study variability were described. Results are presented in evidence tables, with limitations discussed in relation to heterogeneity and reporting quality.

Result Synthesis (Quantitative)

Separate meta-analyses were conducted for RCTs and QSEs. For RCTs, analyses used the *metacont()* function from the *meta* package (version 8.0-2) in R version 4.4.0 (R Foundation).²¹ SMDs were calculated as Cohen *d* with 95% confidence intervals. For QSEs, standardized mean changes were calculated using the standardized mean change using change scores (SMCC) approach, accounting for the paired pre-post structure.²² These were computed with *escalc()* from the *metafor* package (version 4.8-0) in R and meta-analyzed with *rma()*.

SMD and SMCC values of 0.2, 0.5, and 0.8 were interpreted as small, medium, and large in magnitude, respectively.²³ Subgroup analyses were conducted based on prespecified rationales (eTable 4 in Supplement 1). A 2-sided $P \leq .05$ indicated heterogeneity between subgroups. Given the prespecified nature and expected variation in true associations, random-effects meta-analyses were applied, estimating between-study variance separately within each subgroup for RCTs and QSEs. Heterogeneity was assessed with the I^2 statistic (low, 0%-40%; moderate, 30%-60%; substantial, 50%-90%; considerable, 75%-100%).²³ Forest plots were

Figure 1. PRISMA 2020 Flow Diagram



generated to visualize pooled estimates and confidence intervals. Separate analyses were performed for each mental health outcome domain.

Results

A total of 1815 records were identified through database searches. After removing 353 duplicates, 1462 records were screened, of which 47 studies met the inclusion criteria and were included in the final synthesis. Additionally, 3 studies were identified through manual reference list screening, bringing the total number of included studies to 50 (Figure 1). The extracted data for included studies are provided in eTable 1 in Supplement 1; for registered trials, see eTable 2 in Supplement 1.

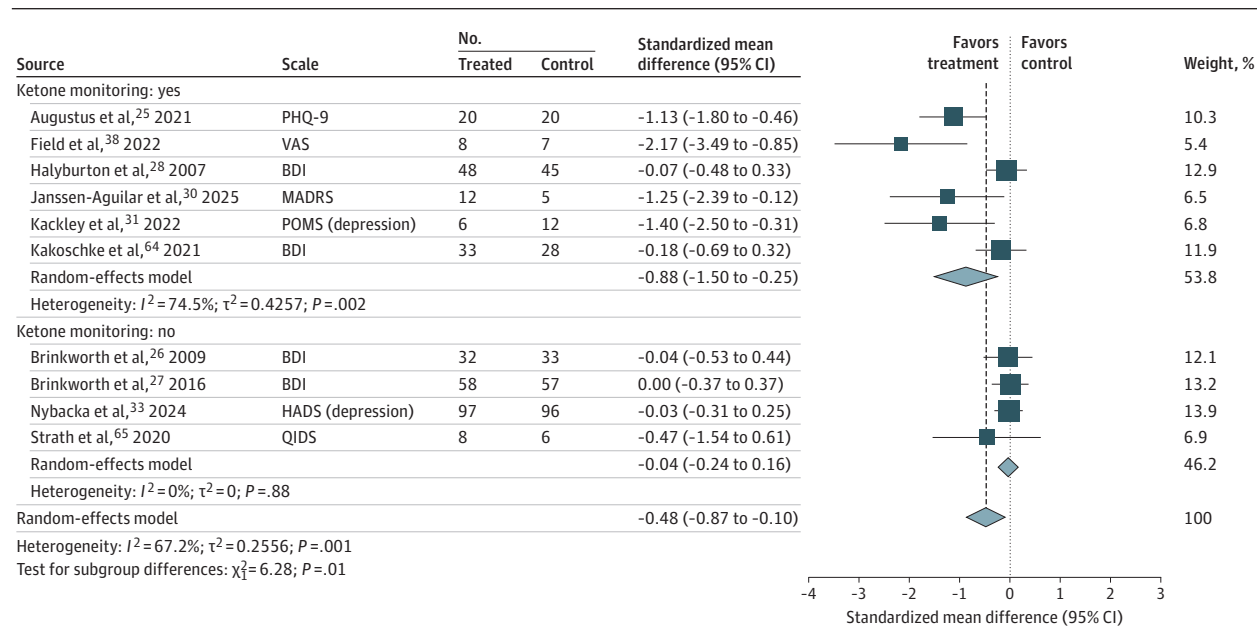
Quality Assessment

All included articles were appraised using the appropriate JBI critical appraisal tools: 14 with the RCT tool, 17 with the QSE study tool, 5 with the analytical CS tool, 6 with the case series tool, and 8 with the case report tool. Each study was evaluated based on its design to assess internal validity, reporting clarity, and potential bias. Nonapplicable criteria (eg, randomization, blinding) were marked as not assessed (detailed results in eTable 3 in Supplement 1).

Study Characteristics

A total of 50 studies (41 718 participants, mean ages 18-70 years) from 15 countries were included, most from the US ($n = 23$), and were primarily QSE studies ($n = 17$) or RCTs ($n = 14$). Studies were published between 1965 and 2025, with output rising from 2019 and peaking in 2024. Sex distribution was re-

Figure 2. Forest Plot Summarizing the Associations of Ketogenic Diets on Depressive Symptoms in Randomized Clinical Trials, Stratified by Ketone Monitoring Status



BDI indicates Beck Depression Inventory; HADS, Hospital Anxiety and Depression Scale; MADRS, Montgomery-Åsberg Depression Rating Scale; PHQ-9, Patient Health Questionnaire-9; POMS, Profile of Mood States; QIDS, Quick Inventory of Depressive Symptomatology; VAS, Visual Analog Scale.

ported in 48 studies (41 651 participants), among whom 22 658 participants (54.3%) were female. Female representation ranged from 0% to 100%, with 18 studies reporting exclusively female samples.

Depressive symptoms were the most frequently evaluated domain ($n = 41$),²⁴⁻⁶² followed by anxiety symptoms ($n = 29$).^{24, 26-28, 30, 31, 33, 37-44, 46-48, 50, 51, 53-55, 57, 59, 62-65} A total of 22 studies enrolled participants with formal psychiatric diagnoses, including BD^{40, 42, 52, 53, 62} and MDD,^{51, 52, 54, 61} while 33 studies focused on nonpsychiatric populations with metabolic, neurological, or general health conditions.^{24-39, 41, 43-47, 49, 50, 56, 57, 59, 63-67}

Additional Results

Detailed descriptions of KD interventions, adverse events, and qualitative findings from RCTs, QSEs, CS studies, case series, and case reports are provided in the eResults in Supplement 1.

Quantitative Synthesis Results

RCTS

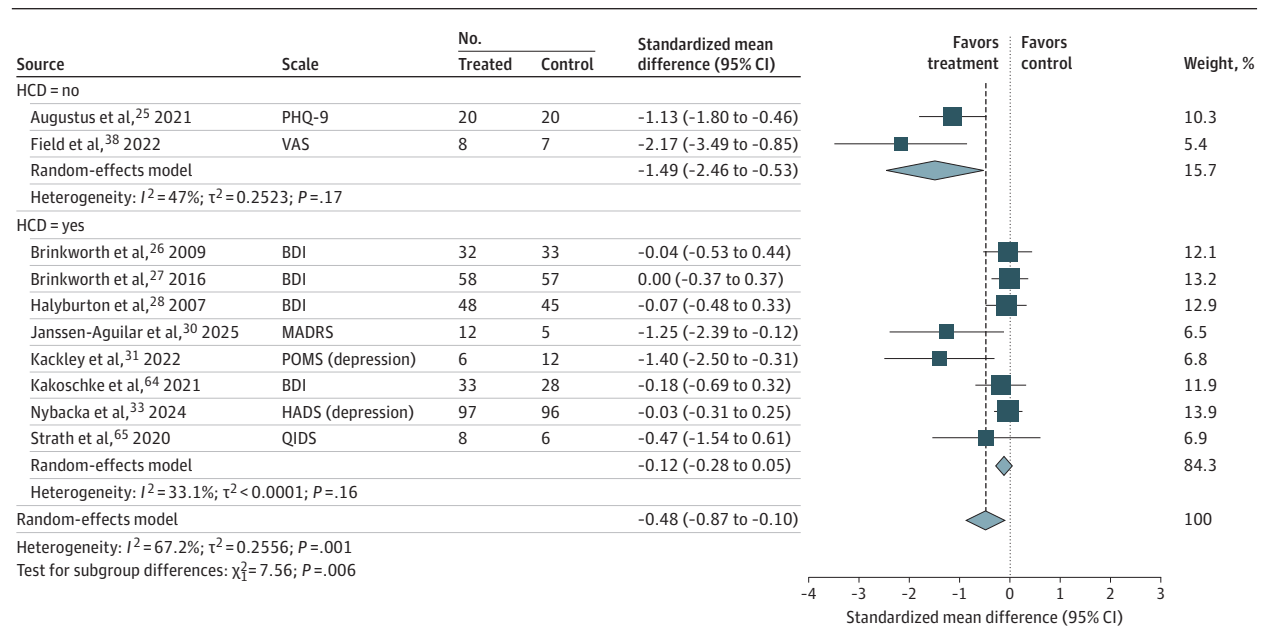
Ten RCTs ($n = 631$) evaluated KD associations with depressive symptoms (Figure 2). The pooled SMD was -0.48 (95% CI, -0.87 to -0.10 ; $I^2 = 67.2\%$), indicating a small to medium benefit compared with control diets. Without ketone monitoring, the association was small and nonsignificant (SMD, -0.04 ; 95% CI, -0.24 to 0.16 ; $I^2 = 0\%$), whereas with monitoring, it was large and significant (SMD, -0.88 ; 95% CI, -1.50 to -0.25 ; $I^2 = 74.5\%$). The difference between subgroups with vs without ketone monitoring was -0.84 (95% CI,

-1.50 to -0.18 ; $P = .01$), suggesting ketone monitoring may moderate the association.

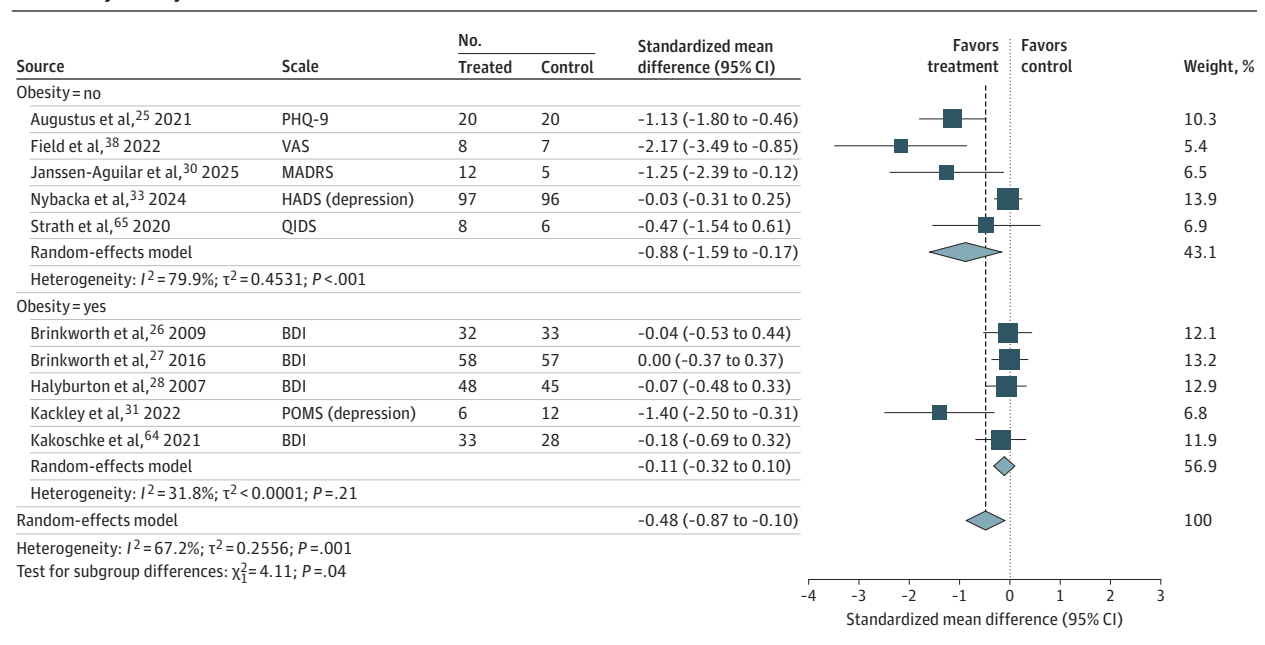
When stratified by control diet, low-fat diet presence (yes vs no) did not yield significant subgroup differences (eFigure 1 in Supplement 1). In contrast, high-carbohydrate diet (HCD) presence showed significant differences (difference in SMD between without and with HCD, -1.37 ; 95% CI, -2.35 to -0.39 ; $P = .006$), with heterogeneity reduced from 67.2% overall to 47% and 33.1% within subgroups without and with HCD, respectively (Figure 3). Studies without HCD showed a large, significant association (SMD, -1.49 ; 95% CI, -2.46 to -0.53 ; $I^2 = 47\%$), whereas those with HCD showed no significant association.

Subgroup analysis by obesity status (obese vs nonobese) also indicated significant differences (difference in SMD between nonobese and obese, -0.77 ; 95% CI, -1.51 to -0.03 ; $P = .04$) (Figure 4). Among nonobese participants, the association was large and significant, although accompanied by substantial heterogeneity (SMD, -0.88 ; 95% CI, -1.59 to -0.17 ; $I^2 = 79.9\%$). In contrast, among obese participants, the association was small and not statistically significant (SMD, -0.11 ; 95% CI, -0.32 to 0.10 ; $I^2 = 31.8\%$).

When stratified by diet classification, significant subgroup differences were observed between very low-carbohydrate diets (VLCD) and low-carbohydrate diets (LCD) (difference in SMD between VLCD and LCD, -0.75 ; 95% CI, -1.35 to -0.15 ; $P = .01$) (Figure 5). Only the VLCD subgroup demonstrated a significant, large association (SMD, -0.79 ; 95% CI, -1.36 to -0.23), with substantial heterogeneity ($I^2 = 71.9\%$).

Figure 3. Forest Plot Summarizing the Associations of Ketogenic Diets on Depressive Symptoms in Randomized Clinical Trials, Stratified by High-Carbohydrate Comparator Diet (HCD) Status

BDI indicates Beck Depression Inventory; HADS, Hospital Anxiety and Depression Scale; MADRS, Montgomery-Åsberg Depression Rating Scale; PHQ-9, Patient Health Questionnaire-9; POMS, Profile of Mood States; QIDS, Quick Inventory of Depressive Symptomatology; VAS, Visual Analog Scale.

Figure 4. Forest Plot Summarizing the Associations of Ketogenic Diets on Depressive Symptoms in Randomized Clinical Trials, Stratified by Obesity Status

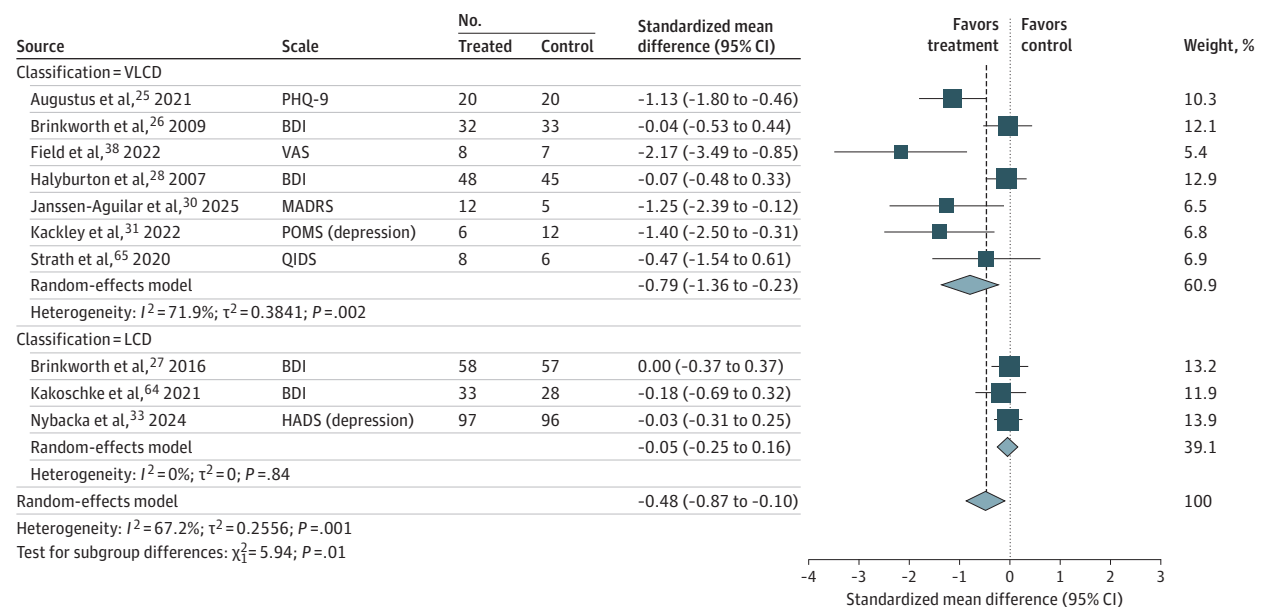
BDI indicates Beck Depression Inventory; HADS, Hospital Anxiety and Depression Scale; MADRS, Montgomery-Åsberg Depression Rating Scale; PHQ-9, Patient Health Questionnaire-9; POMS, Profile of Mood States; QIDS, Quick Inventory of Depressive Symptomatology; VAS, Visual Analog Scale.

Finally, when stratified by intervention duration (4-6 weeks, 8-9 weeks, and ≥ 10 weeks), no significant subgroup differences were observed (eFigure 2 in Supplement 1).

Nine RCTs (with 672 participants) evaluating the associations of KDs with anxiety symptoms were in-

cluded (eFigure 3 in Supplement 1). The pooled SMD was -0.03 (95% CI, -0.18 to 0.12 ; $I^2 = 41\%$), which did not reach statistical significance. Subgroup analyses based on ketone monitoring did not show any differences in outcomes.

Figure 5. Forest Plot Summarizing the Associations of Ketogenic Diets on Depressive Symptoms in Randomized Clinical Trials, Stratified by Carbohydrate-Restriction Category of the Intervention (Very Low-Carbohydrate Diet [VLCD] vs Low-Carbohydrate Diet [LCD])



LCDs defined as 11%-20% of energy from carbohydrates; VLCD defined as $\leq 10\%$ of energy from carbohydrates. BDI indicates Beck Depression Inventory; HADS, Hospital Anxiety and Depression Scale; MADRS, Montgomery-Åsberg

Depression Rating Scale; PHQ-9, Patient Health Questionnaire-9; POMS, Profile of Mood States; QIDS, Quick Inventory of Depressive Symptomatology; VAS, Visual Analog Scale.

Similarly, stratification by low-fat diet (yes vs no) showed no significant subgroup differences (eFigure 4 in Supplement 1). In contrast, HCDs (yes, $n = 8$; no, $n = 1$) yielded significant differences and reduced heterogeneity from 41% overall to 0% within subgroups, although neither subgroup showed a significant association (eFigure 5 in Supplement 1).

Subgroup analyses by obesity status (obese vs non-obese), diet classification (VLCD vs LCD), and intervention duration (4-6 weeks, 8-9 weeks, and ≥ 10 weeks) did not reveal any significant subgroup differences (eFigures 6-8 in Supplement 1, respectively).

QSEs

Nine QSEs evaluated depressive symptoms, and 6 assessed anxiety after KD interventions (eFigure 9 in Supplement 1). For depression, the pooled SMCC was -0.66 (95% CI, -0.83 to -0.50 ; $I^2 = 0\%$). Studies with ketone monitoring showed a consistent medium association (SMCC, -0.62 ; 95% CI, -0.80 to -0.43 ; $I^2 = 0\%$), whereas those without monitoring had a larger association (SMCC, -0.88 ; 95% CI, -1.51 to -0.25 ; $I^2 = 60.5\%$), although subgroup differences were not significant. No significant subgroup differences were observed for neurological disorder status, other comorbidities, or intervention duration (eFigures 10-12 in Supplement 1, respectively).

For anxiety, the pooled SMCC was -0.58 (95% CI, -0.81 to -0.36 ; $I^2 = 0\%$) (eFigure 9 in Supplement 1), with no significant subgroup associations for ketone monitoring, neurological disorder, comorbidities, or duration (eFigures 13-16 in Supplement 1, respectively).

Discussion

This systematic review and meta-analysis found that KD interventions were associated with small to moderate improvements in depressive symptoms in RCT but with no significant changes in anxiety. Greater improvements in depressive symptoms were observed with ketone monitoring, very low-carbohydrate interventions, absence of high-carbohydrate comparators, and in nonobese participants. No consistent moderators were identified for anxiety.

QSE findings were consistent with RCTs, indicating associations between KD interventions and improvements in depressive symptoms. Subgroup analyses identified no significant moderators, with associations generally consistent regardless of ketone monitoring, neurological status, comorbidities, or intervention duration. Slightly larger improvements were observed without ketone monitoring, although these were not statistically significant. Anxiety symptoms were also associated with improvements, with results consistent across participant and intervention characteristics.

These results suggest that KDs may benefit both depression and anxiety across diverse populations, including MDD, GAD, posttraumatic stress disorder (PTSD), and SZ, as well as nonpsychiatric groups. Depressive and anxiety symptoms were the most frequently assessed outcomes, and most studies reported reductions in severity. Stronger and more consistent associations in QSEs may reflect greater adherence and en-

agement in open-label settings, although lower internal validity warrants caution.

These meta-analytic estimates combine heterogeneous populations, designs, and protocols and should not be interpreted as uniform or as evidence of causality. Associations with KD response appear to vary and may be influenced by diagnosis, baseline severity, metabolic phenotype, adherence, and comorbidity; depressive symptoms showed more consistent associations than anxiety. Nutritional ketosis may be associated with benefit in some contexts, although available data do not indicate a linear association between higher BHB levels and greater symptom reduction. These pooled estimates summarize current evidence and highlight priorities for future research rather than establish generalizable conclusions.

The most consistent findings were for depressive symptoms, with improvements reported in randomized and non-randomized trials, case series, and case reports.^{25-28,30,31,33-36,38,39,41-43,46,51-55,57-62,64,67-69} Anxiety symptoms also showed significant improvement.^{24,26,28,30,33,37-39,43-46,51,53-55,57,59,60,63} Improvements in psychotic symptoms and mood stabilization were noted in individuals with SZ or BD following dietary ketosis.^{40,42,52,53,55,62,68-70} One feasibility trial in patients with PTSD also showed meaningful symptom reductions,⁷¹ although additional trials in trauma-related populations are needed.

Although RCTs offer greater internal validity and QSEs may yield larger estimates due to nonspecific within-person change, the RCT-QSE differences also appear related to intervention fidelity. In RCTs, biochemical verification of nutritional ketosis was associated with clearer antidepressant outcomes; for anxiety, the pattern was similar but underpowered. Verification was more frequent in QSEs (depression: 7 of 9 [78%]; anxiety: 5 of 6 [83%]) than in RCTs (depression: 6 of 10 [60%]; anxiety: 6 of 9 [67%]). Heterogeneity was absent in RCTs without monitoring ($I^2 = 0\%$) but substantial in those with monitoring ($I^2 = 74.6\%$), indicating that monitoring alone did not account for variability. Several large RCTs did not consistently verify ketosis and/or used higher-carbohydrate comparators with pragmatic support, aligning with smaller observed associations. Larger QSE estimates—particularly for depression—may reflect both design-related bias and closer adherence to a ketogenic state. Given the evidence hierarchy, RCTs provide the most reliable estimates; within this stratum, associations with anxiety remain inconclusive.

The association between ketone monitoring and larger RCT associations is most plausibly explained by enhanced treatment fidelity rather than bias alone. Objective monitoring confirms exposure, reinforces adherence, enables timely troubleshooting, and reduces exposure misclassification by capturing both intentional and unintentional nonadherence, common in free-living settings and otherwise biasing estimates toward the null. However, monitoring may covary with factors that could inflate observed associations, such as greater participant-clinician contact, feedback-related measurement reactivity, and unequal dietary counseling intensity. Most outcomes were self-reported, introducing the potential for performance and detection bias, as well as differential attrition when ketosis was not achieved. Ketone monitoring

was therefore interpreted as a proxy for metabolic engagement, while recognizing that residual bias cannot be excluded.

The biological plausibility of KDs for psychiatric disorders is supported by converging evidence from molecular, cellular, and clinical domains. Ketone bodies, such as BHB, have been proposed to confer neuroprotective properties through pathways including inhibition of histone deacetylases, activation of Akt-RhoGTPase signaling, and promotion of microglial ramification, which have been linked to antidepressant-like and neuroplastic phenotypes in preclinical models.^{8,10} KDs enhance mitochondrial bioenergetics, reduce reactive oxygen species, and upregulate antioxidant enzymes, aligning with oxidative phosphorylation deficits observed in mood disorders.^{4,5} Ketosis also triggers a hormetic cellular response via activation of Nrf2, sirtuins, and AMP-activated protein kinase, increasing resilience to oxidative and inflammatory stress.⁴ Neurotransmitter balance is further restored by increasing GABAergic tone and reducing glutamatergic excitotoxicity, mechanisms relevant in BD and SZ.^{7,9} Because these neurobiological mechanisms are thought to require sustained exposure to circulating ketone bodies, metabolic factors that limit ketosis may be associated with reduced clinical response.⁷² Consistent with this, this study's RCT subgroup analysis found a large association with antidepressant outcomes in nonobese participants, but a negligible association in those with obesity. Nutritional ketosis is more readily achieved in individuals without obesity, as higher insulin sensitivity and more efficient lipid oxidation promote lipolysis and hepatic ketone production.⁷² In contrast, obesity is associated with insulin resistance, compensatory hyperinsulinemia, and reduced metabolic flexibility, which can delay or blunt the ketogenic response.⁷²

Beyond proposed neurobiological mechanisms, KDs may be associated with restoration of metabolic integrity in psychiatric populations with comorbid metabolic dysfunction. They may bypass cerebral glucose hypometabolism—reported in major psychiatric illnesses—by providing alternative energy substrates and stabilizing neural networks.^{7,9} Modulation of gut microbiota may also affect neuroinflammation, neurotransmission, and brain-gut signaling.^{8,11} Improvements in insulin sensitivity, reduced glucose variability, and cardiometabolic markers further support KDs' potential relevance in psychiatric care.^{12,16} These mechanisms provide a rationale for evaluating KDs as adjunctive therapies in MDD, GAD, BD, and SZ.

However, heterogeneity across studies limits generalizability. Interventions differed in composition and delivery, and ketosis verification ranged from biochemical confirmation to self-report, with biomarkers measured via urine, blood, breath, or combinations of these. The presence or absence of dietary counseling or structured support may also have influenced adherence and outcomes.

Blinding is inherently difficult in KD studies, as participants are often aware of what they are eating. Most included studies did not use blinding, and in the single-blinded trial,³³ the adequacy of blinding—particularly for participants—was unclear. None of the included studies used formal tools to assess blinding integrity or participant expectancy, making it

difficult to determine whether observed associations were influenced by demand characteristics or placebo responses. Addressing these methodological limitations in future trials—such as by incorporating expectancy measures or assessing the success of blinding—could strengthen confidence in the findings.

The safety profile of KD appeared favorable. Commonly reported symptoms—constipation, fatigue, headache, nausea, and diarrhea—were consistent with those described during ketogenic adaptation. Less commonly reported symptoms, including dizziness, cravings, increased thirst, irritability, and sleep disturbances, were observed mainly in case series and reports. One serious adverse event, euglycemic ketoacidosis in a participant receiving sodium-glucose cotransporter 2 inhibitors, was reported. Adverse event reporting was more frequent with ketone monitoring, suggesting enhanced surveillance or closer oversight.

Future research should prioritize large-scale RCTs using standardized ketogenic protocols, consistent ketone monitoring, and stratification by psychiatric diagnosis and metabolic phenotype. Long-term follow-up and adherence tracking are essential to assess sustainability across diverse populations. Comparative trials against established psychiatric and lifestyle interventions (eg, antidepressants, Mediterranean diet) are needed to contextualize efficacy. Mechanistic studies should examine neurobiological pathways, including mitochondrial function, neurotransmission, insulin signaling, and inflammation, with biomarker integration (eg, BHB, C-reactive protein, neuroimaging, transcriptomic or metabolomic profiling) to identify predictors of treatment response.⁷³

Strengths and Limitations

This review has several strengths. To our knowledge, it is the first to systematically evaluate the psychiatric associations of KD and LCDs using both qualitative synthesis and meta-analysis. Strengths include a comprehensive search strategy, large pooled sample size, and inclusion of diverse psychiatric diagnoses and dietary protocols. Stratifying results by study

design and symptom domain provided more granular insights into which populations and outcomes may benefit most.

Nonetheless, important limitations must be acknowledged. Studies varied in design quality, sample size, intervention length, adherence verification, and outcome measures. Heterogeneous clinical populations, baseline depression severity, and potential KD mechanisms may introduce variability not fully addressed in the primary analysis. Although most studies assessed depressive symptoms and fewer examined anxiety, differences in participant characteristics and comorbidities may limit generalizability. Most studies were small, and adverse event and dropout reporting were inconsistent, constraining safety conclusions. While our findings suggest a potential association between KDs and improvements in depressive and anxiety symptoms, they do not establish causality. Further large, high-quality trials are needed, and the long-term sustainability and real-world applicability of KDs in psychiatric populations remain uncertain.

Conclusions

Our findings in this systematic review and meta-analysis suggest that KDs may confer therapeutic benefits for depressive symptoms, whereas evidence for anxiety remains preliminary. Discrepancies between RCTs and QSEs likely reflect design differences and intervention fidelity. Improvements have been reported across varied populations, and adverse events were generally mild; however, these findings should not be interpreted as uniform or causal, and not all patients are likely to benefit. Generalizability is limited by heterogeneity in protocols, comparator diets, adherence support, outcome measures, and study quality, as well as the short follow-up in many trials. Future work should prioritize large, preregistered RCTs with standardized KD protocols, rigorous ketone monitoring, appropriate comparators, structured dietary support, extended follow-up, and predefined psychiatric outcomes to determine who benefits and under what conditions.

ARTICLE INFORMATION

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